

DBMS – ORDER AND ORDER ITEMS MANAGEMENT

Project Title: Lenskart E-commerce Database Management System

1. Database Design

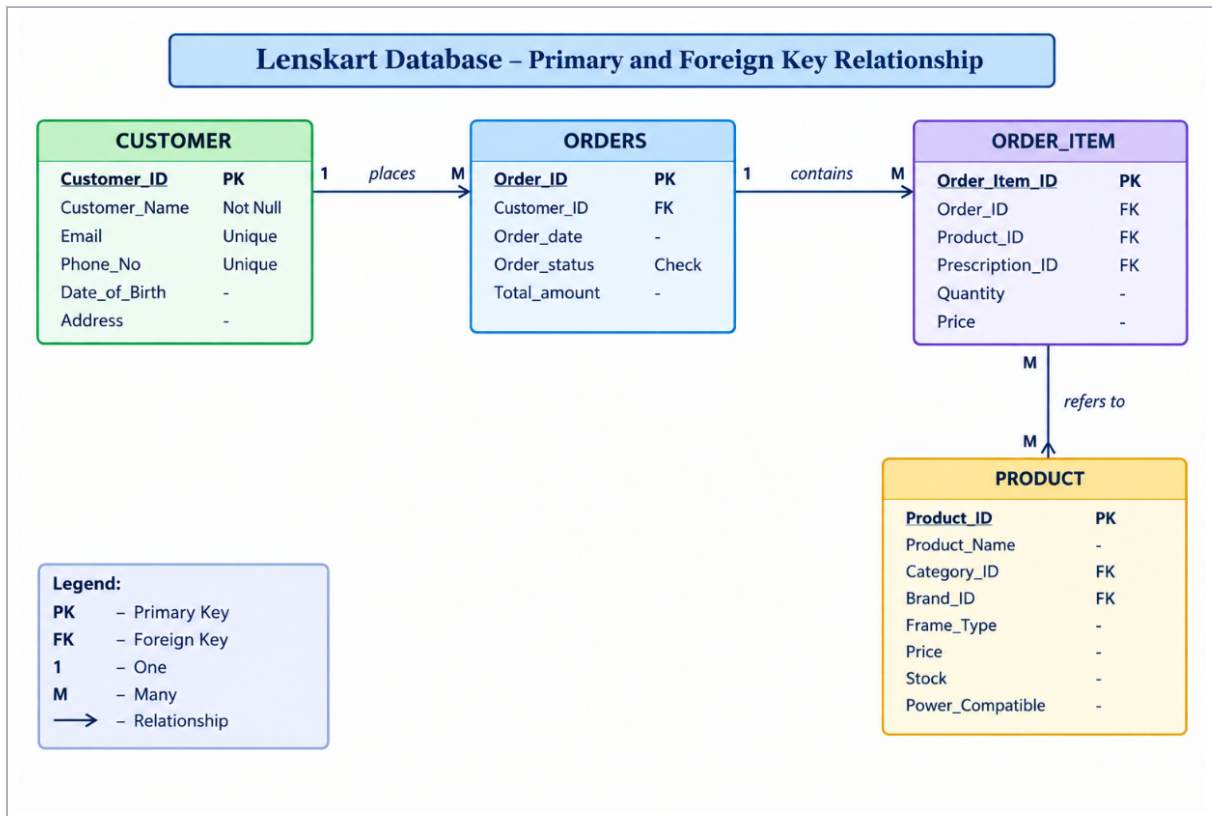
1.1 Order Table

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_date	-
Order_status	Check
Total_amount	-

1.2 Order_items Table

Attribute	Key Type
Order_item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Prescription_ID	Foreign Key
Quantity	-
Price	-
Sub_total	-

2. Primary and Foreign Key Relationship



3. Table Creation

3.1 Create Orders Table

```
CREATE TABLE Orders (  
    Order_ID NUMBER PRIMARY KEY,  
    Customer_ID NUMBER,  
    Order_Date DATE,  
    Order_Status VARCHAR2(20),  
    Total_Amount NUMBER(10,2),  
    CONSTRAINT fk_order_customer  
        FOREIGN KEY (Customer_ID)  
        REFERENCES Customer(Customer_ID),  
    CONSTRAINT chk_order_status  
        CHECK (Order_Status IN  
            ('Pending', 'Processing', 'Shipped', 'Delivered', 'Cancelled'))  
);
```

3.2 Create Order_Item Table

```
CREATE TABLE Order_Item (  
    Order_Item_ID NUMBER PRIMARY KEY,  
    Order_ID NUMBER,  
    Product_ID NUMBER,  
    Prescription_ID NUMBER,  
    Quantity NUMBER NOT NULL,  
    Price NUMBER(10,2) NOT NULL,  
    CONSTRAINT fk_orderitem_order  
        FOREIGN KEY (Order_ID)  
        REFERENCES Orders(Order_ID),  
    CONSTRAINT fk_orderitem_product  
        FOREIGN KEY (Product_ID)  
        REFERENCES Product(Product_ID),  
    CONSTRAINT fk_orderitem_prescription  
        FOREIGN KEY (Prescription_ID)  
        REFERENCES Prescription(Prescription_ID)  
);
```

4. Data Insertion

4.1 Orders Records

```
INSERT INTO Orders  
VALUES (101, 1, TO_DATE('20-09-2026','DD-MM-YYYY'),  
        'Pending', 2499);  
1 row created.  
  
INSERT INTO Orders  
VALUES (102, 2, TO_DATE('21-09-2026','DD-MM-YYYY'),  
        'Processing', 1899);  
1 row created.  
  
INSERT INTO Orders  
VALUES (103, 3, TO_DATE('22-09-2026','DD-MM-YYYY'),  
        'Shipped', 2999);  
1 row created.  
  
INSERT INTO Orders  
VALUES (104, 4, TO_DATE('23-09-2026','DD-MM-YYYY'),  
        'Delivered', 1599);  
1 row created.  
  
INSERT INTO Orders  
VALUES (105, 5, TO_DATE('24-09-2026','DD-MM-YYYY'),  
        'Pending', 3499);  
1 row created.  
  
INSERT INTO Orders  
VALUES (106, 6, TO_DATE('25-09-2026','DD-MM-YYYY'),  
        'Processing', 1299);
```

```

1 row created.

INSERT INTO Orders
VALUES (107, 7, TO_DATE('26-09-2026','DD-MM-YYYY'),
       'Shipped', 2199);
1 row created.

INSERT INTO Orders
VALUES (108, 8, TO_DATE('27-09-2026','DD-MM-YYYY'),
       'Delivered', 1799);
1 row created.

INSERT INTO Orders
VALUES (109, 9, TO_DATE('28-09-2026','DD-MM-YYYY'),
       'Pending', 2699);
1 row created.

INSERT INTO Orders
VALUES (110, 10, TO_DATE('29-09-2026','DD-MM-YYYY'),
       'Delivered', 3999);
1 row created.

```

4.2 Order_Item Records

```

INSERT INTO Order_Item VALUES (1, 101, 201, 301, 1, 1200);
1 row created.

INSERT INTO Order_Item VALUES (2, 102, 202, 302, 1, 1500);
1 row created.

INSERT INTO Order_Item VALUES (3, 103, 203, 303, 2, 800);
1 row created.

INSERT INTO Order_Item VALUES (4, 104, 204, 304, 1, 1000);
1 row created.

INSERT INTO Order_Item VALUES (5, 105, 205, 305, 1, 1300);
1 row created.

INSERT INTO Order_Item VALUES (6, 106, 201, 306, 2, 1200);
1 row created.

INSERT INTO Order_Item VALUES (7, 107, 202, 307, 1, 1500);
1 row created.

INSERT INTO Order_Item VALUES (8, 108, 203, 308, 3, 800);
1 row created.

INSERT INTO Order_Item VALUES (9, 109, 204, 309, 1, 1000);
1 row created.

INSERT INTO Order_Item VALUES (10, 110, 205, 310, 2, 1300);
1 row created.

```

5. Data Retrieval

5.1 Display Orders

```
SELECT * FROM Orders;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	ORDER_STATUS	TOTAL_AMOUNT
101	1	20-SEP-26	Pending	2499
102	2	21-SEP-26	Processing	1899
103	3	22-SEP-26	Shipped	2999
104	4	23-SEP-26	Delivered	1599
105	5	24-SEP-26	Pending	3499
106	6	25-SEP-26	Processing	1299
107	7	26-SEP-26	Shipped	2199
108	8	27-SEP-26	Delivered	1799
109	9	28-SEP-26	Pending	2699
110	10	29-SEP-26	Delivered	3999

```
10 rows selected.
```

5.2 Display Order_Items

```
SELECT * FROM Order_Item;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	PRESCRIPTION_ID	QUANTITY	PRICE
1	101	201	301	2	1200
2	102	202	302	1	1500
3	103	203	303	2	800
4	104	204	304	1	1000
5	105	205	305	1	1300
6	106	201	306	2	1200
7	107	202	307	1	1500
8	108	203	308	3	800
9	109	204	309	1	1000
10	110	205	310	2	1300

```
10 rows selected.
```

6. Order and Order_Items Modification

6.1 Update Order Status

```
UPDATE Orders
SET Order_Status = 'Delivered'
WHERE Order_ID = 101;
```

1 row updated.

```
UPDATE Orders
SET Order_Status = 'Delivered'
WHERE Order_ID IN (105,106,107);
```

3 rows updated.

```
select * from orders;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	ORDER_STATUS	TOTAL_AMOUNT
101	1	20-SEP-26	Delivered	2499
102	2	21-SEP-26	Processing	1899
103	3	22-SEP-26	Shipped	2999
104	4	23-SEP-26	Delivered	1599
105	5	24-SEP-26	Delivered	3499
106	6	25-SEP-26	Delivered	1299
107	7	26-SEP-26	Delivered	2199
108	8	27-SEP-26	Delivered	1799
109	9	28-SEP-26	Pending	2699
110	10	29-SEP-26	Delivered	3999

10 rows selected.

6.2 Update Order Quantity

```
UPDATE Order_Item
SET Quantity = 2
WHERE Order_Item_ID = 1;
```

1 row updated.

```
SELECT *
FROM Order_Item
WHERE Order_Item_ID = 1;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	PRESCRIPTION_ID	QUANTITY	PRICE
1	101	201	301	2	1200

```
UPDATE Order_Item
SET Quantity = 2
WHERE Order_Item_ID = 5;
```

```
SELECT *
FROM Order_Item
WHERE Order_Item_ID IN (1,2,5);
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	PRESCRIPTION_ID	QUANTITY	PRICE
1	101	201	301	2	1200
2	102	202	302	1	1500
5	105	205	305	2	1300

7. Customer – Order History Report

```
SELECT
    c.CUSTOMER_ID,
    c.CUSTOMER_NAME,
    o.ORDER_ID,
    o.ORDER_DATE,
    p.PRODUCT_ID,
    p.PRODUCT_NAME,
    oi.QUANTITY,
    oi.PRICE,
    (oi.QUANTITY * oi.PRICE) AS SUBTOTAL
FROM Customer c
JOIN Orders o
    ON c.CUSTOMER_ID = o.CUSTOMER_ID
JOIN Order_Item oi
    ON o.ORDER_ID = oi.ORDER_ID
JOIN Product p
    ON oi.PRODUCT_ID = p.PRODUCT_ID
ORDER BY c.CUSTOMER_ID, o.ORDER_DATE;
```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
ORDER_DAT	PRODUCT_ID	
PRODUCT_NAME		
QUANTITY	PRICE	SUBTOTAL
1	Arun Kumar	101
20-SEP-26	201	
Classic Eyeglasses		
2	1200	2400
CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
ORDER_DAT	PRODUCT_ID	
PRODUCT_NAME		
QUANTITY	PRICE	SUBTOTAL
2	Priya	102
21-SEP-26	202	
Stylish Sunglasses		
1	1500	1500

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
ORDER_DAT	PRODUCT_ID	
PRODUCT_NAME		
QUANTITY	PRICE	SUBTOTAL
3	Karthik	103
22-SEP-26	203	
Reading Glasses		
2	800	1600
CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID
ORDER_DAT	PRODUCT_ID	
PRODUCT_NAME		
QUANTITY	PRICE	SUBTOTAL
4	Divya	104
23-SEP-26	204	
Contact Lenses		
1	1000	1000

CUSTOMER_ID		CUSTOMER_NAME		ORDER_ID

ORDER_DAT		PRODUCT_ID		

PRODUCT_NAME				

QUANTITY		PRICE	SUBTOTAL	

		5	Vignesh	105
24-SEP-26		205		
Blue Light Glasses				
		1	1300	1300

CUSTOMER_ID		CUSTOMER_NAME		ORDER_ID

ORDER_DAT		PRODUCT_ID		

PRODUCT_NAME				

QUANTITY		PRICE	SUBTOTAL	

		6	Keerthana	106
25-SEP-26		201		
Classic Eyeglasses				
		2	1200	2400

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID
```

```
ORDER_DAT PRODUCT_ID
```

```
PRODUCT_NAME
```

```
QUANTITY PRICE SUBTOTAL
```

```
7 Rahul 107
```

```
26-SEP-26 202
```

```
Stylish Sunglasses
```

```
1 1500 1500
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID
```

```
ORDER_DAT PRODUCT_ID
```

```
PRODUCT_NAME
```

```
QUANTITY PRICE SUBTOTAL
```

```
8 Nandhini 108
```

```
27-SEP-26 203
```

```
Reading Glasses
```

```
3 800 2400
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID
```

```
ORDER_DAT PRODUCT_ID
```

```
PRODUCT_NAME
```

```
QUANTITY PRICE SUBTOTAL
```

```
9 Sanjay 109
```

```
28-SEP-26 204
```

```
Contact Lenses
```

```
1 1000 1000
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID
```

```
ORDER_DAT PRODUCT_ID
```

```
PRODUCT_NAME
```

```
QUANTITY PRICE SUBTOTAL
```

```
10 Harini 110
```

```
29-SEP-26 205
```

```
Blue Light Glasses
```

```
2 1300 2600
```

```
10 rows selected.
```

8. Customer – wise Order Summary

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    COUNT(o.Order_ID) AS Total_Orders,
    SUM(o.Total_Amount) AS Total_Amount
FROM Customer c
JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
GROUP BY
    c.Customer_ID,
    c.Customer_Name
ORDER BY
    c.Customer_ID;
```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS
1	Arun Kumar	1
2499		
2	Priya	1
1899		
3	Karthik	1
2999		

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS
4	Divya	1
1599		
5	Vignesh	1
3499		
6	Keerthana	1
1299		

```
CUSTOMER_ID CUSTOMER_NAME TOTAL_ORDERS
```

```
TOTAL_AMOUNT
```

```
-----  
7 Rahul 1  
2199  
8 Nandhini 1  
1799  
9 Sanjay 1  
2699
```

```
CUSTOMER_ID CUSTOMER_NAME TOTAL_ORDERS
```

```
TOTAL_AMOUNT
```

```
-----  
10 Harini 1  
3999
```

```
10 rows selected.
```